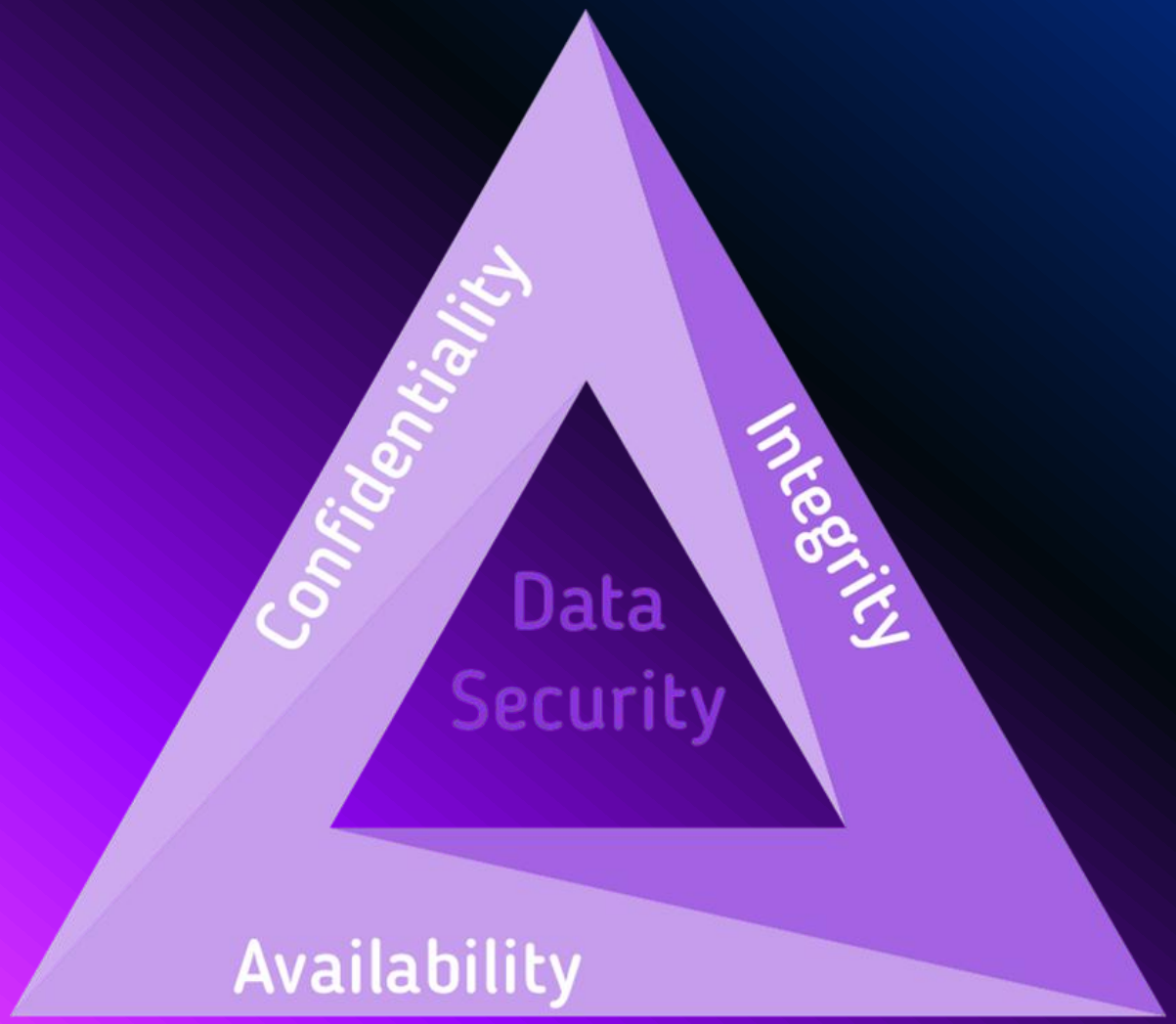
MiniSploit:

A CVE SCANNER +
PROOF OF VULNERABILITY TOOL



Why is Vulnerability Scanning Important?

- Identifies security weaknesses
- Protects Confidentiality, Integrity, and Availability
- Prevents exploitation

Related Tools

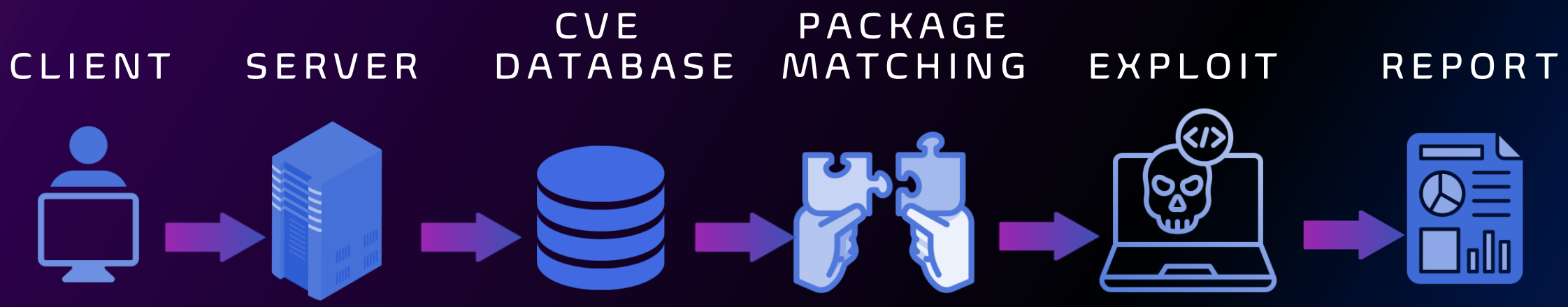
- Common Vulnerabilities and Exploitation (CVE) Databases:
  - NVD (National Vulnerability Database)
  - Snyk Security Database
- Vulnerability Scanners:
  - Nessus Vulnerability Scanner
  - OpenVAS (Open Vulnerability Assessment System)
- Exploitation Resources:
  - Metasploit
  - Exploit-DB

Problem & Goal

- **Problem:**
  - Tools are fragmented
  - Limited visibility into the process and impact
- **Goal of MiniSploit:**
  - Combine data, detection, exploitation, and reporting
  - Provide unified and transparent system

How Does MiniSploit Work?

System Overview

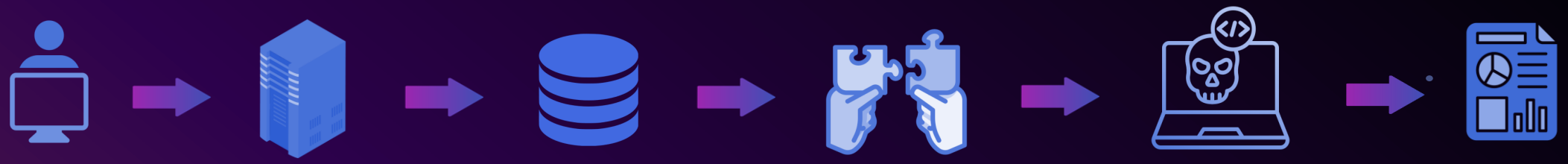


Components

- **MiniSploit Server (Ubuntu VM)**
  - Local CVE database pulled from the Ubuntu CVE Tracker.
  - Listens for client scan requests
  - Compares package versions against CVE database
  - Returns vulnerability matches to client

Ubuntu Docker Images (Clients)

- Intentionally outdated packages
- Sends OS version and installed package list to the server
- Receives CVE matches from the server
- Prompts to run proof of concept exploit(s)
- Generates security report



DEMO

MiniSploit-Server (March28) [Running] - Oracle VirtualBox

File Machine View Input Devices Help

Activities Terminal Apr 9 15:18

Home

mick@MiniSploit-Server: ~/minisploit

```
mick@MiniSploit-Server:~/minisploit$ python3 -m server
```

mick@MiniSploit-Server: ~/minisploit/docker

```
mick@MiniSploit-Server:~/minisploit/docker$ docker-compose up -d --build
```

Right Ctrl

Challenges

- Setting up reproducible vulnerable Docker environments
- Handling discrepancies between package names
  - (ex. inetutils-telnet vs inetutils)
- Handling package versions that are not strictly sequential

`sudo`	`1.9.15p5-3ubuntu5`	`1.9.15p5-3ubuntu5.24.04.1`

Limitations & Future Work

- Limited to Ubuntu CVE data
  - Future: Expand to other OS
- Package matching may miss edge cases
  - Future: Improve comparison logic
- Limited number of PoC exploits
  - Future: Expand exploit library / automate discovery

Conclusion

- MiniSploit:
  - Demonstrates full vulnerability workflow
  - Connects data, detection, exploitation, and reporting
  - Highlights real-world security impact

Thank You

Mickaela

Questions?